Computer Science (083)

MASTER QUESTION BANK • CLASS 12 BOARD EXAMS

Unit 1: Programming & Data Structures

Strategy Tip: Python Functions & Scope

Questions on functions typically test parameter passing (positional, default, keyword), variable scope (Local vs. Global, the `global` keyword), and output generation. Always trace variable values step-by-step in rough work.

Q1: Output & Scope Resolution — 2 Marks

Find and write the output of the following Python code:

```
def Change(P, Q=30): P = P + Q Q = P - Q print(P, "#", Q) return P R = 150 S = 100
R = Change(R, S) print(R, "#", S) S = Change(S)
```

Expected Answer:
250 # 150
250 # 100
130 # 100

Q2: Random Module (Expected Output) — 2 Marks

What are the possible outcomes executed from the following code? Also specify the maximum and minimum values that can be assigned to variable N.

```
import random Colors = ["RED", "BLUE", "GREEN", "YELLOW", "CYAN", "MAGENTA"] N =
random.randint(1, 3) for i in range(N, 5): print(Colors[i], end="*")
```

Expected Answer:
Minimum value of N: 1, Maximum value of N: 3.
Possible outputs include: `BLUE*GREEN*YELLOW*CYAN*` (if N=1), `GREEN*YELLOW*CYAN*` (if N=2), or `YELLOW*CYAN*` (if N=3).

Q3: Data Structures (Stack Implementation) — 3 Marks

Write Python functions `Push(Book)` and `Pop(Book)` to add a new book and delete a book from a List of books acting as a Stack. A book is represented as a list `[BookNo, BookName]`.

```
def Push(Stack): bno = int(input("Enter Book No: ")) bname = input("Enter Book
Name: ") Stack.append([bno, bname]) print("Book added successfully.") def Pop(Stack): if
len(Stack) == 0: # Underflow condition print("Stack is empty.") else: deleted_book =
Stack.pop() print("Deleted book:", deleted_book)
```

Unit 1 Continued: File Handling

Quick Reference: File Modes

Mode	Purpose	File Pointer Position	Creates File if Not Exist?
`r` / `rb`	Read-only (Text/Binary)	Beginning of file	No (raises Error)
`w` / `wb`	Write (Text/Binary)	Beginning (Truncates/Overwrites)	Yes
`a` / `ab`	Append (Text/Binary)	End of file	Yes
`r+`	Read & Write	Beginning of file	No

Q4: Text File Handling (Word/Line Count)

3 Marks

Write a function `count_words()` to read a text file "STORY.TXT" and display the total number of lines, and the number of words starting with the letter 'A' or 'a'.

```
def count_words(): try: with open("STORY.TXT", "r") as file: lines = file.readlines() line_count = len(lines) a_words = 0 for line in lines: words = line.split() for w in words: if w[0].lower() == 'a': a_words += 1 print(f"Total Lines: {line_count}") print(f"Words starting with 'A': {a_words}") except FileNotFoundError: print("File not found.")
```

Q5: Binary File Handling (Pickle Module)

4 Marks

A binary file "STUDENT.DAT" contains records as dictionaries: `{"RollNo":1, "Name":"Amit", "Marks":85}`. Write a function `Search(R)` that takes a RollNo and displays the corresponding record. If not found, display "Record Not Found".

```
import pickle def Search(R): found = False try: with open("STUDENT.DAT", "rb") as f: while True: try: record = pickle.load(f) if record["RollNo"] == R: print(record) found = True except EOFError: break except FileNotFoundError: print("Database does not exist.") if not found: print("Record Not Found")
```

Q6: CSV File Handling

3 Marks

Write a Python program to write multiple lists of user-entered data (EmpNo, EmpName, Salary) into a CSV file named "EMPLOYEE.csv" until the user enters 'n' to stop.

```
import csv def write_csv(): with open("EMPLOYEE.csv", "w", newline="") as f: writer = csv.writer(f) writer.writerow(["EmpNo", "EmpName", "Salary"]) # Header ans = 'y' while ans.lower() == 'y': eno = int(input("Enter Employee No: ")) ename = input("Enter Name: ") sal = float(input("Enter Salary: ")) writer.writerow([eno, ename, sal]) ans = input("Add more? (y/n): ")
```

Unit 3: Database Management & SQL

Consider the following tables **PRODUCT** and **CLIENT**. Write SQL queries for Q7(a) to Q7(d).

Table: PRODUCT			
P_ID	**PName**	**Price**	**Stock**
P01	Mouse	500	50
P02	Keyboard	1200	30
P03	Monitor	8000	15
P04	Webcam	2500	40

Table: CLIENT		
C_ID	**CName**	**P_ID**
C101	Ajay	P02
C102	Vijay	P04
C103	Rahul	P02
C104	Sneha	P01

Q7: SQL Query Execution — 4 Marks

- **(a)** To display the details of products whose price is between 1000 and 3000.
  `SELECT * FROM PRODUCT WHERE Price BETWEEN 1000 AND 3000;`
- **(b)** To display the Product Name and Price in descending order of Price.
  `SELECT PName, Price FROM PRODUCT ORDER BY Price DESC;`
- **(c)** To display the Client Name (CName) and corresponding Product Name (PName) they purchased (using JOIN).
  `SELECT C.CName, P.PName FROM CLIENT C, PRODUCT P WHERE C.P_ID = P.P_ID;`
- **(d)** To count how many clients have purchased Product 'P02'.
  `SELECT COUNT(*) FROM CLIENT WHERE P_ID = 'P02';`

Key Database Terminologies

DDL (Data Definition Language): Commands used to define or alter table structures (e.g., `CREATE`, `ALTER`, `DROP`).
DML (Data Manipulation Language): Commands used to manipulate data within tables (e.g., `INSERT`, `UPDATE`, `DELETE`).
Primary Key: A column (or set of columns) that uniquely identifies each row in a table. It cannot contain NULL values.

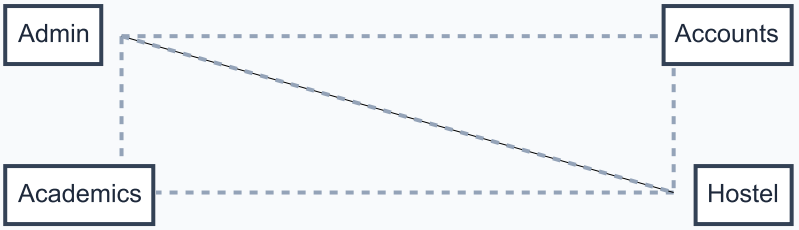
Q8: Python - SQL Interface — 4 Marks

Write a Python code to connect to a MySQL database named `SchoolDB` and fetch all records from the `STUDENT` table where marks are greater than 90.

```
import mysql.connector try: # Establish connection con = mysql.connector.connect( host="localhost", user="root", password="password123", database="SchoolDB" ) cursor = con.cursor() # Execute Query query = "SELECT * FROM STUDENT WHERE Marks > 90" cursor.execute(query) # Fetch and display records data = cursor.fetchall() for row in data: print(row) except mysql.connector.Error as err: print("Database error:", err) finally: if con.is_connected(): cursor.close() con.close()
```

Unit 2: Computer Networks (Case Study)

Q9: Global EduTech is setting up its new campus in Delhi. They have 4 main buildings: **Admin**, **Accounts**, **Academics**, and **Hostel**. As a network expert, answer the queries below based on the given distances and number of computers.



Distance between buildings	
Admin to Accounts	50 m
Admin to Academics	150 m
Admin to Hostel	200 m
Accounts to Academics	110 m
Academics to Hostel	60 m

Number of Computers	
Admin	120
Accounts	40
Academics	150
Hostel	20

Case Study Queries

- **(a) Suggest the most suitable building to house the main server of this campus. Justify your answer.**
  Answer: **Academics**. Following the 80-20 rule, the building with the maximum number of computers (150) should host the server to minimize network traffic and cable load.
- **(b) Suggest the placement of a Repeater in the network.**
  Answer: A repeater should be placed between **Admin and Academics** (150m) and **Admin and Hostel** (200m). Repeaters regenerate signals over long cables, typically required when distance exceeds 100 meters (for twisted pair ethernet).
- **(c) Which device should be installed in each building to connect all the computers together?**
  Answer: **Switch** (or Hub). A switch is preferred as it is an intelligent device that filters and forwards packets to the specific destination.
- **(d) Suggest the best wired medium and best network topology to efficiently connect these buildings.**
  Answer: Medium: **Optical Fiber Cable (OFC)** (for high bandwidth and less attenuation). Topology: **Star Topology**, with Academics as the central node.

Exam Tip: Common Network Abbreviations

Memorize these frequently asked expansions:
• **VoIP:** Voice over Internet Protocol
• **TCP/IP:** Transmission Control Protocol / Internet Protocol
• **HTTP:** HyperText Transfer Protocol
• **MAC:** Media Access Control (Physical Address of NIC)
• **SMTP:** Simple Mail Transfer Protocol